When F1 Fails: Granularity-Aware Evaluation for Dialogue Topic Segmentation

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Preprint. Under review.

Abstract

Dialogue topic segmentation supports summarization, retrieval, memory management, and conversational continuity. Despite decades of prior work, evaluation practice in dialogue topic segmentation remains dominated by strict boundary matching and F1-based metrics [1, 2], even as modern LLM-based conversational systems increasingly rely on segmentation to manage conversation history beyond the model's fixed context window, where unstructured context accumulation degrades efficiency and coherence.

This paper introduces an evaluation objective for dialogue topic segmentation that treats boundary density and segment coherence as primary criteria, alongside tolerance-based boundary accuracy. Through extensive cross-dataset empirical evaluation, we show that reported performance differences across dialogue segmentation benchmarks are driven not by model quality, but by mismatches in annotation granularity and sparsely annotated topic boundaries. This indicates that many reported improvements arise from evaluation artifacts rather than improved boundary detection.

We evaluated multiple, structurally distinct dialogue segmentation strategies across eight dialogue datasets spanning task-oriented, open-domain, meeting-style, and synthetic interactions. Across these settings, we observe high segment coherence combined with extreme oversegmentation relative to sparse labels, producing misleadingly low exact-match F1 scores. We show that topic segmentation is best understood as selecting an appropriate granularity rather than predicting a single correct boundary set. We also propose a topic segmentation algorithm that separates boundary scoring from boundary selection, and demonstrate that this separation makes boundary density a controllable design choice and clarifies why thresholds fail to transfer across datasets.

1 Introduction

Dialogue topic segmentation is commonly framed as identifying points in a conversation where "the subject changes." This framing is appealingly simple. It is also incorrect. Topic structure is often gradual, nested, and perspective-dependent.

Despite this, many dialogue segmentation benchmarks treat annotated boundaries as objective ground truth and evaluate systems using strict boundary matching and F1-based metrics [1]. These metrics collapse distinct failure modes into a single number. Small boundary shifts are often harmless when the goal is coherent topic blocks, while missing or adding a few boundaries can be catastrophic if boundaries drive downstream actions such as summarizing or dropping earlier turns.

This paper addresses that mismatch by changing what we report and what we optimize. We define an evaluation objective that jointly reports (i) tolerant boundary accuracy, (ii) boundary density, and (iii) segment coherence. We also propose an algorithmic separation between boundary scoring and boundary selection that makes boundary density explicit and controllable. Together, these contributions let us distinguish detection failures from granularity mismatches (i.e., discrepancies between the segmentation resolution at which boundaries are produced and that implied by the annotation scheme) and explain why a single global threshold fails to transfer across datasets.

We emphasize, however, that boundary-based metrics such as F1 are appropriate when annotation granularity is consistent between training, evaluation, and intended use. The failure modes illustrated here arise when segmentation granularity differs across datasets or between system behavior and annotation intent, which is increasingly common in dialogue benchmarks and downstream applications.

We do not propose a new segmentation model. Our contribution is an evaluation perspective and a principled separation between boundary scoring and boundary selection that exposes granularity effects obscured by standard metrics. Concretely, we make the following contributions:

- We demonstrate that boundary-based metrics (including F1 and W-F1) can be dominated by boundary density alignment: even non-semantic baselines achieve competitive scores when their output density matches the gold annotation.
- We propose an evaluation objective that reports boundary density and segment coherence alongside tolerant boundary accuracy, exposing failure modes hidden by standard boundary metrics.
- We introduce and empirically validate a topic segmentation framework that separates boundary scoring from boundary selection, making boundary density an explicit design choice rather than an accidental side effect of threshold tuning.

1.1 Motivation: Conversational Memory in AI Chat Systems

Large language model (LLM) based chat systems operate under finite context windows, which require practical systems to manage conversational history explicitly. As conversations grow longer, earlier turns must be dropped, summarized, or selectively retrieved. Topic segmentation decisions therefore directly determine which portions of prior dialogue are preserved, compressed, or reintroduced.

Importantly, this challenge does not disappear as context windows grow larger. Unstructured accumulation of earlier dialogue increases computational cost and dilutes relevance, reducing the model's ability to focus on salient information. Effective conversational systems therefore require explicit structure over dialogue history rather than unbounded context expansion.

In this setting, topic segmentation functions as a memory control mechanism rather than a purely linguistic task. Poor segmentation can cause relevant context to be discarded prematurely or irrelevant context to be retained, leading to degraded long-horizon coherence and unstable behavior across turns. The absence of robust episodic memory remains a major limitation of current LLM-based systems, particularly in multi-session interactions.

This work is motivated by the development of *Episodic*, a conversational memory system that organizes dialogue into topic-structured representations to support persistent interactions with LLM-based assistants [3]. Within such systems, boundary placement governs summarization, retrieval, and context reintegration, making segmentation granularity a first-order systems design decision rather than an evaluation detail.

2 Background and Related Work

Topic segmentation has a long history in text processing, with evaluation practice, dataset design, and granularity assumptions evolving somewhat independently. We organize prior work along these three dimensions.

2.1 Boundary-Based Evaluation Metrics

Topic segmentation evaluation has historically centered on boundary alignment. Early metrics such as Pk [4] penalized segmentations where predicted and gold boundaries fell in different segments under a sliding window. WindowDiff [5] refined this approach to address Pk's known biases. Both metrics assume a single correct segmentation and measure deviation from it. More recent dialogue segmentation work has largely adopted precision, recall, and exact-match F1 computed over boundary positions [1, 2]. These metrics inherit the same assumption: that annotated boundaries represent ground truth rather than one valid segmentation among many. When annotation granularity varies across datasets or differs from system behavior, this assumption produces misleading evaluations.

2.2 Dialogue Segmentation Datasets

Dialogue topic segmentation introduces complexity beyond written text due to turn-taking, discourse phenomena, and ill-defined topic boundaries. Several benchmarks have emerged with distinct annotation philosophies. DialSeg711 [6] and SuperDialseg [7] provide densely annotated dialogue corpora, while TIAGE [8] annotates topic boundaries in task-oriented dialogue with sparser labels tied to task structure. MultiWOZ [9] provides domain annotations that can be interpreted as coarse topic boundaries. Open-domain and semi-structured datasets such as DailyDialog [10], Taskmaster [11], Topical-Chat [12], and QMSum [13] reflect further variation in annotation intent, from synthetic concatenation boundaries to summarization-driven segmentation. This diversity in annotation philosophy produces systematic differences in boundary density across datasets—differences that standard metrics do not distinguish from detection quality.

2.3 Segmentation Granularity

Classic segmentation methods such as TextTiling [14] and Choi-style benchmarks [15] established the paradigm of predicting a single boundary set evaluated against gold annotations. Subsequent work has refined scoring models and architectures, but the assumption that a single correct granularity exists has remained largely unexamined. Prior work rarely treats segmentation granularity as a first-class, controllable variable. Boundary density is typically a side effect of threshold tuning rather than an explicit design choice, and evaluation protocols do not distinguish between detection failures (placing boundaries at wrong locations) and granularity mismatches (producing a different number of boundaries than the annotation scheme implies). This paper addresses that gap by separating boundary scoring from boundary selection and proposing evaluation criteria that make granularity effects explicit.

3 Problem Setting

Topic boundaries are not intrinsic properties of conversations. They depend on segmentation granularity, task requirements, and annotation intent. A model that produces fine-grained discourse-phase boundaries can be operationally reasonable yet score poorly against a dataset that annotates only coarse task transitions. Conversely, a model tuned to sparse labels can suppress meaningful structure.

This paper treats topic segmentation as two separate operations:

1. **Boundary scoring:** assign a score to each candidate boundary position.
2. **Boundary selection:** choose which candidates become output boundaries, controlling boundary density and spacing.

Collapsing these operations makes boundary density an accidental byproduct of threshold tuning. Our baseline experiments confirm that this produces systematically misleading results: the effect is demonstrated in Section 6.1.

4 Datasets

We evaluated multiple, structurally distinct dialogue segmentation strategies across eight dialogue datasets spanning diverse dialogue regimes. The purpose of cross-dataset evaluation here is not only to compare performance, but to expose how annotation philosophy and boundary density interact with evaluation metrics.

Task-oriented dialogue datasets include SuperDialseg [7], DialSeg711 [6], TIAGE [8], and MultiWOZ [9]. Open-domain and semi-structured datasets include DailyDialog (synthetic concatenations)[10], Taskmaster[11], Topical-Chat [12], and QMSum [13]. Several datasets provide sparse boundaries by design or annotate boundaries with intent (e.g., summarization units), which is precisely the setting where exact-match F1 becomes fragile.

5 Methodology

Our approach outputs a score for each canonical between-message position indicating the likelihood of a topic boundary. We map all gold and predicted boundaries to canonical between-message indices to avoid speaker-based off-by-one artifacts.

The key methodological choice is to separate boundary scoring from boundary selection. Scoring aims to generalize: it should rank likely topic transitions above unlikely ones. Selection is a decision rule: it chooses boundary density and spacing to match an annotation regime or an application requirement. In deployed use, selection may be implemented as a commitment policy, which is a decision rule that accumulates evidence over time and determines when to emit a boundary, rather than emitting boundaries directly from the scoring model.

5.1 Topic Segmentation Algorithm

Given a dialogue consisting of messages $m_1, \ldots, m_T$, we define candidate boundary positions $i \in \{1, \ldots, T-1\}$ between adjacent messages.

Boundary scoring. For each candidate position i, a base scorer computes a real-valued score $s_i \in \mathbb{R}$, where larger values indicate stronger evidence of a topic boundary. Scores are computed using a window-vs-window comparison: a window of turns preceding position i is compared against a window of turns following i, rather than relying on the adjacent message pair alone.

Commit gating. We distinguish between *boundary scoring* and *boundary commitment*. The scorer ranks candidate boundary positions, while *commit gating* determines when a boundary is actually emitted. Commit gating is an online decision mechanism that accumulates evidence over time and emits a boundary only when commitment criteria are satisfied (e.g., confidence, persistence, or rate constraints). We use "commit gating" and "commitment policy" interchangeably to refer to the same boundary selection mechanism. This separation decouples boundary selection from the granularity of the underlying scoring function.

Boundary selection (rule used in experiments). In our experiments, commit gating reduces to a static selection rule based on thresholding with minimum spacing.

1. Select all candidate positions i such that $s_i \geq \tau$.

2. Enforce a minimum spacing g via greedy non-maximum suppression: candidates are processed in descending score order, and a candidate i is accepted only if no previously accepted boundary lies within g positions.

The resulting set of topic boundaries is $B \subseteq \{1, \ldots, T-1\}$.

For Figure 2, we allow the threshold τ to vary over time (logged as `min_evidence`); all other aspects of the selection rule remain unchanged. During evidence accumulation, candidate scores are evaluated against a fixed pre-change reference. Freezing both the pre-boundary context and the boundary-adjacent anchor turn ensures that the policy measures persistence of topic departure rather than repeated detection of the same topic transition.

5.2 Scoring Model Architecture

The boundary scoring model is a fine-tuned DistilBERT-based binary classifier. For each candidate boundary position i, the model encodes a left context window and a right context window around i using the input format `[CLS] pre-window [SEP] post-window [SEP]`. Unless otherwise specified, the scoring model uses a symmetric local context window of k=4 utterances before and after each candidate boundary (total window size 9), truncated at dialogue boundaries when insufficient context is available. The final hidden state of the `[CLS]` token is passed through a linear projection layer to produce a scalar score $s_i \in \mathbb{R}$ representing boundary confidence.

The model is trained using binary cross-entropy loss on boundary labels. No architectural modifications beyond the classification head are introduced; the purpose of this implementation is to provide a stable scoring signal rather than to optimize segmentation performance.

5.3 Training Procedure

We train the scoring model in three stages:

Stage 1: Synthetic Pretraining. We construct synthetic splice examples by concatenating real dialogue segments and labeling the splice point as a boundary. We remove boundary-local artifacts (e.g., greetings) to prevent spurious cues, and we evaluate a negative control set of single-segment examples to verify that the model does not hallucinate boundaries.

Stage 2: Supervised Fine-Tuning on Benchmarks. We next evaluate the effect of supervised fine-tuning on benchmark datasets. Table 7 summarizes performance over five epochs.

Stage 3: Temperature Calibration. We apply temperature scaling to calibrate score magnitudes by optimizing a single scalar temperature $T > 0$ on a held-out validation set (minimizing negative log-likelihood). This operation rescales the logits used to produce boundary scores, improving score comparability and numerical stability across runs.

Importantly, temperature calibration rescales *evidence* but does not determine how many boundaries are produced. Boundary density is controlled exclusively by the selection rule (thresholding and spacing), not by temperature. While changing T shifts the score distribution, it does not encode a target boundary rate and cannot, by itself, enforce consistent segmentation granularity across datasets. Calibration therefore improves the interpretability of scores but does not resolve granularity mismatch, which remains a decision-level issue handled by explicit boundary selection.

Reproducibility. The reference implementation for this work is available in the `paper/` directory of the Episodic repository at `https://github.com/mhcoen/episodic`. This directory contains the

LaTeX source for the paper, all experiment configuration files, evaluation scripts, and code required to reproduce the reported results.

Data availability. All datasets used in this study are publicly available from their original sources and are cited in the paper. Due to licensing restrictions, we do not redistribute raw dialogue data. Instead, the repository provides scripts to download, preprocess, and evaluate each dataset in accordance with its original distribution terms.

6 Evaluation Metrics

Relation to Pk and WindowDiff. Classic segmentation metrics such as Pk and WindowDiff [4, 5] were designed to measure boundary placement accuracy under a single assumed segmentation granularity, and they remain useful for diagnosing alignment errors (e.g., off-by-one boundary shifts) in homogeneous annotation settings. However, like exact-match F1, these metrics implicitly conflate detection quality with granularity alignment: they penalize segmentations that produce additional boundaries even when those boundaries correspond to coherent subtopics omitted by a coarser gold standard.

The evaluation objective proposed here is not a drop-in replacement for Pk or WindowDiff. Instead, it decomposes segmentation behavior into complementary dimensions: tolerant boundary detection (W-F1), boundary density relative to the annotation scheme (BOR), and segment-level coherence (purity and coverage). This decomposition makes it possible to distinguish genuine detection failures from granularity mismatch, a distinction that alignment-based metrics alone cannot express. In practice, Pk or WindowDiff may still be reported for continuity with prior work, but we argue that they should be interpreted alongside explicit boundary density and coherence diagnostics when annotation granularity varies across datasets or applications.

We report three evaluation measures:

- **W-F1:** window-tolerant F1 [5], matching predicted boundaries to gold boundaries within a fixed window.
- **BOR:** boundary oversegmentation ratio, $\mathrm{BOR} = \#B_{\mathrm{pred}}/\#B_{\mathrm{gold}}$.
- **Purity/Coverage:** segment coherence diagnostics from clustering evaluation.

We use a ± 1 message tolerance window for W-F1, following standard practice in dialogue segmentation. We found this window size balances robustness to off-by-one annotation differences against the risk of masking systematic oversegmentation, which is captured separately by BOR.

Segment-level coherence computation. To apply purity and coverage to dialogue segmentation, we treat both predicted and gold segmentations as sequences of contiguous spans between boundaries. Purity is computed by assigning each predicted segment to the gold segment with which it has maximum overlap and measuring the fraction of turns in the predicted segment belonging to that gold segment. Coverage is computed symmetrically by measuring, for each gold segment, the fraction of its turns captured by a single predicted segment. These measures assess whether segments are internally coherent, independent of exact boundary placement within the reference annotation.

Strict exact-match boundary F1 is reported only as a baseline, because it collapses under coarse or sparse annotation and penalizes coherent segmentations at a different granularity. The evaluation objective in this paper treats boundary density (BOR) and segment coherence (purity/coverage) as primary criteria alongside tolerant boundary accuracy (W-F1). This is a genuine limitation of any evaluation tied to a fixed annotation

F1/W-F1	BOR	Purity/Coverage	Interpretation
High	≈ 1	High	Calibrated (aligned granularity)
High	$\gg 1$	High purity	Oversegmentation (coherent fine-grained)
Low	$\gg 1$	High purity	Granularity mismatch (not incoherent)
Low	< 1	High coverage	Undersegmentation (coarser-than-gold)
Low	≈ 1	Low	Detection failure (misplaced/noisy)

Table 1: **Failure modes of boundary-based evaluation.** Distinct segmentation behaviors can collapse to similar boundary scores unless boundary density (BOR) and coherence diagnostics are reported.

scheme: purity and coverage diagnose relative coherence with respect to the reference annotation, but do not provide annotation-independent measures of semantic coherence.

Interpreting metrics through failure modes. W-F1 detects gross misalignment; BOR exposes systematic over- or under-segmentation; purity/coverage indicate whether the resulting segments are coherent and usable. For example, low F1 with high purity and high BOR typically indicates granularity mismatch rather than incoherent segmentation.

Normative implication. When annotation granularity varies across datasets or applications, boundary-based accuracy metrics (including strict F1, W-F1, P_k, and WindowDiff) are not interpretable in isolation. In practical terms, this means that boundary accuracy scores cannot be meaningfully compared across datasets with differing annotation granularity unless boundary density is explicitly controlled or reported. Table 1 summarizes the characteristic failure modes that arise from different combinations of boundary accuracy, boundary density, and segment coherence.

Practical use of the evaluation framework. When evaluating a dialogue segmentation system, we recommend first inspecting BOR to determine whether the system's boundary density aligns with the annotation regime or application requirements. Boundary accuracy metrics (F1 or W-F1) should then be interpreted conditional on this density. Finally, segment coherence diagnostics (purity/coverage) can be used to distinguish coherent fine-grained segmentation from incoherent boundary noise.

Minimum reporting recommendations for dialogue topic segmentation. When reporting or comparing dialogue topic segmentation results across datasets or evaluation settings with differing annotation granularity, we recommend reporting (i) a boundary accuracy metric (e.g., W-F1), (ii) boundary density relative to gold annotations (e.g., BOR), and (iii) at least one segment-level coherence diagnostic.

6.1 Baselines

This subsection examines how simple non-semantic baselines behave under the proposed evaluation objective. The goal is not to establish competitive performance, but to demonstrate how boundary density alone can dominate traditional boundary metrics.

By comparing the proposed model against trivial strategies such as predicting no boundaries, fixed-interval boundaries, or random boundaries matched to the gold density, we isolate which aspects of performance are attributable to semantic boundary detection versus density alignment. Table 2 summarizes these results.

Notably, these baselines highlight a fundamental limitation of boundary-based metrics: when boundary density is matched to the gold annotation, even strategies that ignore the underlying distribution of gold boundaries can achieve deceptively strong F1 and W-F1 scores.

Baseline	Dataset	W-F1	**BOR**	Purity	Coverage
No-boundary	DialSeg711	0.000	**0.00**	0.438	1.000
	SuperSeg	0.113	**0.00**	0.568	1.000
	TIAGE	0.140	**0.00**	0.541	1.000
Every-N	DialSeg711 ($N = 4$)	0.639	**0.97**	0.812	0.823
	SuperSeg ($N = 2$)	0.623	**1.10**	0.830	0.775
	TIAGE ($N = 3$)	0.610	**0.97**	0.849	0.774
Random (BOR=1)	DialSeg711	0.528	**1.00**	0.839	0.833
	SuperSeg	0.678	**1.00**	0.883	0.884
	TIAGE	0.664	**1.00**	0.873	0.869

Table 2: **Baseline segmentation results.** Random and periodic baselines achieve competitive W-F1 and purity when boundary density matches the gold distribution, demonstrating that boundary density alignment alone can produce deceptively strong scores under standard boundary-based metrics.

Method	Dataset	W-F1	**BOR**	F1	Purity	Coverage
TextTiling	DialSeg711	0.648	**1.88**	0.550	0.968	0.746
CSM	DialSeg711	0.412	**0.64**	0.373	0.749	0.921
Random	DialSeg711	0.267	**0.55**	0.098	0.711	0.890
Even	DialSeg711	0.483	**1.00**	0.211	0.806	0.809
TextTiling	TIAGE	0.612	**2.04**	0.445	0.949	0.693
CSM	TIAGE	0.615	**3.97**	0.346	0.976	0.454
Random	TIAGE	0.256	**0.33**	0.121	0.697	0.921
Even	TIAGE	0.620	**1.00**	0.300	0.832	0.838

Table 3: **External dialogue segmentation methods re-evaluated under the proposed granularity-aware framework.** Boundary density (BOR) varies substantially across methods and datasets, producing distinct segmentation regimes despite comparable boundary accuracy scores.

7 Results

This section evaluates the proposed separation between boundary scoring and boundary selection under differing annotation granularity. The results below instantiate multiple failure modes summarized in Table 1, including for previously published methods re-evaluated under the same framework.

7.1 External Methods Under Granularity-Aware Evaluation

To assess whether the granularity effects identified in this work are specific to our scoring model or reflect broader evaluation behavior, we re-evaluated a representative subset of published dialogue segmentation methods under the proposed granularity-aware framework. The selected methods include simple baselines (random and periodic segmentation), a classical unsupervised approach (TextTiling), and a coherence-based model (CSM). All methods were evaluated using the same boundary canonicalization and metrics as in previous sections.

Several patterns in Table 3 align with the failure modes summarized in Section 6. Simple periodic segmentation achieves competitive boundary accuracy when its output density aligns with the gold annotation (BOR $\approx$ 1), despite lacking semantic grounding. Classical methods such as TextTiling exhibit recall-

Method	Dataset	Granularity Regime
TextTiling	DialSeg711	Aggressive (BOR $\gg 1$)
CSM	DialSeg711	Conservative (BOR < 1)
Random	DialSeg711	Conservative (BOR < 1)
Even	DialSeg711	Balanced (BOR ≈ 1)
TextTiling	TIAGE	Aggressive (BOR $\gg 1$)
CSM	TIAGE	Aggressive (BOR $\gg 1$)
Random	TIAGE	Conservative (BOR < 1)
Even	TIAGE	Balanced (BOR ≈ 1)

Table 4: Granularity regimes induced by boundary density under the proposed failure taxonomy. Regime labels are derived directly from BOR thresholds.

Claim	Comparison	Dataset	ΔF1	ΔW-F1	ΔBOR	Relative Granularity Shift
A	TextTiling vs. CSM	DialSeg711	+0.135	+0.220	+2.13	Conservative $\rightarrow$ Aggressive
B	TextTiling vs. CSM	TIAGE	+0.128	+0.219	+1.26	Aggressive $\rightarrow$ More Aggressive
C	TextTiling vs. Even	DialSeg711	+0.277	+0.132	+1.69	Balanced $\rightarrow$ Aggressive

Table 5: **Pairwise audit of runnable SuperDialseg comparisons.** For each claim, positive Δ indicates higher values for the first-listed method. Arrows denote relative changes in boundary density from the second-listed method to the first-listed method, not absolute regime membership. Regime labels follow the thresholds defined in Table 1.

dominated behavior with elevated BOR, yielding high purity alongside aggressive oversegmentation. Notably, the same method (CSM) occupies different granularity regimes across datasets, indicating that apparent performance differences often reflect density mismatch rather than improved boundary detection. For example, the regime shift observed for CSM across datasets is consistent with differences in annotation density rather than changes in boundary detection quality.

For clarity, and following the failure taxonomy in Table 1, we categorize the results in Table 3 as conservative (BOR < 1), balanced (BOR ≈ 1), or aggressive (BOR $\gg 1$), as summarized in Table 4.

Targeted audit of runnable SuperDialseg comparisons. To further examine how reported boundary-metric improvements relate to segmentation granularity, we performed a targeted audit of three runnable comparisons from the SuperDialseg benchmark. Each comparison contrasts methods that differ in reported F1/W-F1 while sharing the same evaluation protocol. These three comparisons exhaust the set of SuperDialseg leaderboard comparisons that can be reproduced using publicly available implementations without retraining or proprietary APIs. We therefore restrict this audit to published methods with deterministic implementations. The InstructGPT baseline reported in SuperDialseg uses `text-davinci-003`, which has since been deprecated and is no longer available; we do not reproduce this baseline.

This behavior aligns with the failure modes summarized in Table 1, where recall-dominated overseg-mentation yields higher boundary accuracy under tolerant metrics despite substantial shifts in segmentation granularity.

7.2 Stage 1: Synthetic Pretraining on Splice Boundaries

This stage isolates whether the scoring model can learn a meaningful boundary signal in a controlled setting, independent of annotation noise. Performance here reflects the model's ability to distinguish genuine

Epoch	F1	W-F1	BOR	Salience	Neg. Ctrl.
1	0.265	0.420	0.31	0.007	0.000
2	0.453	0.688	0.67	0.009	0.032
3	0.489	0.733	0.87	0.010	—

Table 6: **Stage 1 synthetic pretraining on splice boundaries.** Salience denotes the mean boundary score magnitude $\mathbb{E}[|s_i|]$ on the validation set and serves as a diagnostic for score collapse. Negative control (Neg. Ctrl.) reports the predicted boundary rate on single-segment dialogues, which should be near zero; it was not rerun for epoch 3, as the model had already passed this sanity check at epoch 2.

Epoch	Overall F1	W-F1	**BOR**	SuperSeg F1/BOR	TIAGE F1/BOR
1	0.778	0.833	**1.11**	0.788 / 1.13	0.197 / 0.24
2	0.819	0.850	**1.23**	0.830 / 1.25	0.352 / 0.86
3	0.840	0.865	**1.13**	0.852 / 1.14	0.337 / 0.82
4	0.848	0.875	**1.08**	0.862 / 1.08	0.322 / 0.78
5	0.843	0.872	**1.04**	0.857 / 1.05	0.311 / 0.84

Table 7: **Stage 2 supervised fine-tuning on benchmark datasets.** Boundary density converges toward annotated density (BOR ≈ 1) on datasets seen during training, while W-F1 remains high.

topic transitions from non-boundaries before exposure to real dialogue benchmarks. Table 6 summarizes boundary-level behavior during this phase.

7.3 Stage 2: Supervised Fine-Tuning on Benchmarks

This stage evaluates how supervised fine-tuning aligns the scoring model with human annotation conventions on in-distribution datasets. The key question is whether boundary density stabilizes without degrading boundary ranking quality. Table 7 reports performance over five fine-tuning epochs.

7.4 Stage 3: Final Test with Calibration

This stage evaluates generalization under distribution shift, where annotation granularity differs from the training regime. Here, the focus is not peak F1 but the interaction between boundary ranking, boundary density, and calibration.

Final test results are shown in Table 8, which reports evaluation metrics for a representative subset of three datasets spanning sparse, medium, and dense annotation regimes; results on the remaining datasets follow the same qualitative pattern.

Calibration. We learned a single global temperature of $T = 0.976$ for this run. In practice, temperature scaling modestly improves score calibration and threshold stability, but it does not eliminate cross-dataset threshold non-transfer: differences in annotation granularity continue to dominate the relationship between score thresholds and boundary density.

8 Segment Coherence Diagnostics

These coherence diagnostics are essential for interpreting boundary-based metrics under granularity mismatch. In particular, they distinguish between two qualitatively different outcomes that exact-match metrics conflate:

Dataset	W-F1	**BOR**	F1	Pred./Gold Boundaries
DialSeg711	0.767	**2.53**	0.434	5167/2042
SuperSeg	0.584	**0.81**	0.609	2376/2923
TIAGE	0.512	**0.76**	0.368	157/207

Table 8: **Stage 3 final test results.** Boundary density (BOR) is reported alongside boundary accuracy to distinguish granularity mismatch from detection failure.

Model	Dataset	W-F1	**BOR**	Purity	Coverage	F1
Ours	DialSeg711	0.767	**2.53**	0.962	0.651	0.434
	SuperSeg	0.584	**0.81**	0.847	0.915	0.609
	TIAGE	0.512	**0.76**	0.785	0.896	0.368
CSM	DialSeg711	0.413	**0.64**	0.749	0.921	0.373

Table 9: **Segment coherence diagnostics for the proposed model and an external baseline.** High purity with elevated BOR indicates coherent fine-grained segmentation; high coverage with reduced BOR indicates coarser-than-gold segmentation. Both patterns reflect granularity mismatch rather than boundary detection failure.

(1) segmentations that introduce additional but coherent subtopic boundaries, and (2) segmentations that fragment topics into incoherent or mixed segments.

As shown in Table 9, the proposed model maintains high purity across datasets even when boundary density exceeds that of the gold annotation. This indicates that additional predicted boundaries correspond to meaningful conversational structure rather than noise. In contrast, baseline methods with matched boundary density achieve competitive boundary metrics but exhibit lower or unstable coherence, underscoring that density alignment alone does not guarantee usable segmentation.

Taken together, these results demonstrate that segment coherence metrics are necessary to interpret boundary accuracy meaningfully when annotation granularity varies, and that high BOR should not be interpreted as failure in the presence of high purity and coverage.

8.1 Worked Example

We illustrate the effect of annotation granularity on evaluation metrics using a short dialogue example with both coarse gold boundaries and fine-grained model predictions.

This example makes concrete the failure mode identified throughout the paper. The predicted boundaries correspond to coherent conversational subtopics that are omitted by the coarse gold annotation. Under exact-match evaluation, these additional boundaries are treated as false positives, substantially reducing F1 despite correct identification of the major topic transition. The example therefore highlights that the dominant error is one of boundary density rather than boundary placement, and motivates evaluation criteria that distinguish granularity mismatch from detection failure.

9 Discussion

The results demonstrate that boundary scoring generalizes more robustly than boundary selection under heterogeneous annotation granularity. Baseline comparisons in Table 2 expose a central weakness of standard boundary-based metrics: random and periodic strategies achieve competitive W-F1 and purity scores when

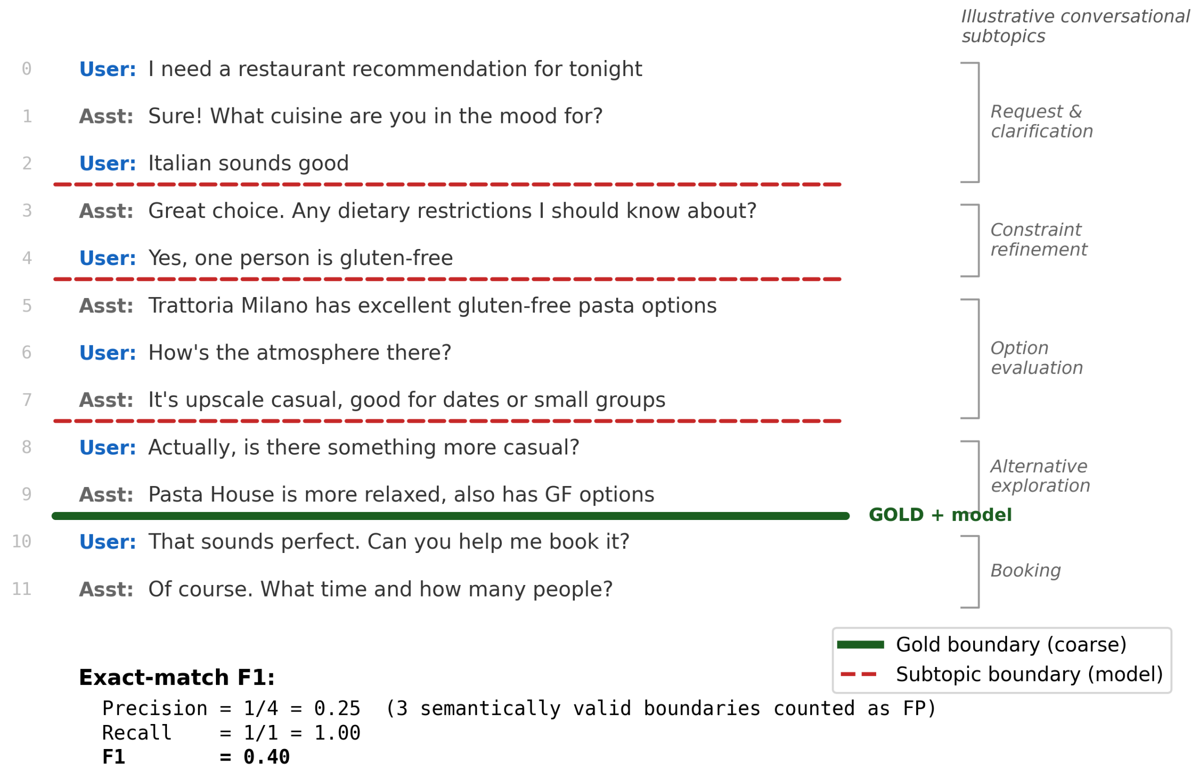

Figure 1: **Granularity mismatch in dialogue topic segmentation.** Gold annotations mark a single coarse topic boundary (browsing → booking), while the model identifies multiple semantically meaningful subtopic transitions. The fine-grained boundaries correspond to coherent conversational phases and are not boundary-positioning errors. Under exact-match evaluation, three valid predicted subtopic boundaries are counted as false positives, yielding F1=0.40 despite correct identification of the coarse transition. The error arises from a mismatch in boundary density and segmentation granularity rather than from incorrect boundary detection.

their output boundary density is matched to the gold annotation, despite lacking any semantic grounding. In contrast, the proposed model yields substantially higher segment purity on datasets with coarse annotations, at the cost of elevated BOR, indicating coherent fine-grained segmentation rather than detection failure.

These findings show that exact-match and tolerance-based boundary metrics conflate segmentation quality with granularity alignment. Importantly, the evaluation failure identified here is not model-specific: any method whose output boundary density is tuned to match the gold annotation can achieve competitive F1 and W-F1 scores, as demonstrated by the non-semantic baselines in Table 2. Adding more segmentation models does not resolve this problem; it only illustrates it further.

Additional experiments with classical change-detection and drift-based strategies further support this conclusion. Such methods (e.g., cumulative-sum and time-aware change detection) often achieve higher W-F1 than the neural scorer by aggressively increasing boundary density, frequently yielding boundary oversegmentation ratios well above 1.0. This demonstrates that strong boundary accuracy under tolerant metrics can be driven by recall-dominated strategies rather than by calibrated segmentation aligned with downstream use. The proposed evaluation objective makes this trade-off explicit: methods with higher W-F1 but elevated BOR are correctly identified as oversegmenting, while more conservative scorers exhibit lower boundary accuracy but better density control.

Across the methods examined in this study, three qualitatively distinct segmentation regimes emerge:

- **Conservative strategies** (e.g., SpeechAct, SummaryProbe) undersegment, producing BOR < 1 and low recall.
- **Balanced strategies** (e.g., Neural(fine), commitment-based selection) achieve BOR ≈ 1 with high segment purity, yielding calibrated segmentation aligned with annotation density.
- **Aggressive strategies** (e.g., CUSUM, time-aware drift detection) maximize recall and W-F1 by substantially increasing boundary density (BOR $\gg 1$), resulting in recall-dominated detection rather than calibrated segmentation.

To confirm that granularity effects are not specific to our scorer, we evaluated CSM [16], a published coherence-based segmentation model, under the same framework. CSM operates in the conservative regime (BOR=0.64), producing fewer boundaries than the gold annotation. Its low strict F1 (0.373) reflects low recall rather than incoherent segmentation: coverage remains high (0.921), indicating that predicted segments are valid but coarser than the reference. This demonstrates that the granularity-F1 interaction identified here generalizes across structurally distinct segmentation methods.

In practice, selection thresholds should therefore be chosen based on intended downstream use rather than optimized solely to match a particular annotation density. Boundary density metrics such as BOR provide a simple diagnostic for tuning selection behavior without retraining the scoring model.

Practical guidance for choosing boundary density. The appropriate boundary selection behavior depends on how segmentation outputs are used downstream. Based on the empirical results in this study, we offer the following high-level guidance:

- **Summarization and context truncation.** When topic boundaries trigger summarization or removal of prior turns, missing a major boundary can be more harmful than introducing extra minor boundaries. Conservative selection is therefore preferable: a lower boundary density (BOR < 1) reduces the risk of prematurely discarding relevant context, even if some distinct topics are merged.
- **Retrieval and navigation.** For applications that rely on topic boundaries to index or retrieve relevant conversation segments, finer-grained segmentation can be beneficial provided segments remain coherent. In this regime, higher boundary density (BOR > 1) is acceptable as long as purity remains high, indicating that additional boundaries correspond to meaningful subtopics rather than noise.
- **Exploratory analysis and annotation.** When segmentation is used for qualitative analysis, annotation support, or exploratory inspection, fine-grained boundaries are often desirable. In such settings, boundary density may substantially exceed that of existing annotations, and evaluation should emphasize segment coherence over exact boundary alignment.

These guidelines reinforce the central claim of this paper: boundary density is not an intrinsic property of a conversation, but a design choice that should be tuned to application requirements rather than optimized to match a given annotation scheme. This dependence on selection behavior motivates treating boundary selection as a distinct, explicit step.

9.1 Boundary Selection as a Separate Step

Separating boundary scoring from boundary selection makes boundary density an explicit control parameter rather than an incidental consequence of threshold tuning.

Figure 2 illustrates the consequence of this separation: different scoring granularities produce substantially different candidate boundary rates, yet an explicit commitment policy enforces stable output boundary distributions that are invariant to the scorer's granularity. Crucially, this invariance does not arise from score rescaling or post-hoc normalization. The scoring function is unchanged; identical output distributions emerge solely from the selection mechanism's evidence accumulation and commit gating behavior. This behavior illustrates that separating scoring from selection enables boundary density to be controlled independently of scoring granularity, motivating its treatment as a design parameter rather than an evaluation artifact.

9.2 Computational Cost

Boundary selection is computationally negligible: it operates on a vector of scores and applies thresholding and minimum spacing. Boundary scoring dominates cost because it requires running the neural model over candidate positions (window-vs-window inputs around each position). The separation therefore has practical value in deployed systems: selection can be retuned per dataset or application without recomputing scores, while recomputing scores requires model inference. Evidence accumulation within a commitment policy is likewise lightweight relative to scoring, since it reuses the fixed-reference scores and maintains only decision state.

9.3 On Dataset Choice and External Validity

Public dialogue datasets are necessarily imperfect proxies for real human–AI interactions. While several conversational AI datasets have been released, we are not aware of any publicly available corpus of long, open-ended human–AI conversations with turn-level topic boundary annotations suitable for evaluating segmentation granularity. Our claims therefore concern evaluation behavior under heterogeneous annotation schemes rather than direct modeling of user behavior. The consistency of failure modes across diverse datasets indicates that these effects arise from evaluation practice rather than dataset-specific artifacts.

10 Limitations and Future Work

This work does not introduce a new neural backbone for dialogue topic segmentation. Instead, it contributes (i) an evaluation objective that makes boundary density and segment coherence explicit and (ii) a topic segmentation algorithm that separates boundary scoring from boundary selection. We do not include a direct ablation comparing synthetic pretraining against training from scratch. Preliminary experiments indicate that splice-boundary pretraining primarily stabilizes early training dynamics rather than improving final performance, but a systematic ablation remains an important direction for future work.

Purity and coverage are computed with respect to the same gold annotations whose granularity we argue is often mismatched to model behavior. Accordingly, these measures should not be interpreted as independent estimates of semantic coherence. Rather, they function as relative diagnostics: high purity rules out incoherent mixing of gold topics, while elevated BOR distinguishes fine-grained subdivision from boundary noise. Independent coherence measures (e.g., embedding-based similarity or human judgments) would be valuable complements, but are not required for the present goal of diagnosing evaluation failure modes under mismatched annotation granularity.

Statistical variability. We do not report confidence intervals or significance tests for the metrics presented in this work. Our primary claims concern qualitative failure modes and regime-level behavior (e.g., recall-dominated oversegmentation versus calibrated segmentation), which manifest as large and consistent differences in boundary density and coherence across datasets and methods. In preliminary experiments, variance across random seeds and training runs was small relative to these effects and did not alter the qualitative conclusions. Future work could include multi-run statistical reporting for finer-grained comparisons within a given segmentation regime, but such analysis is not required to support the central evaluation claims made here.

Inter-annotator agreement (IAA) statistics could also provide additional empirical support for the claim that topic boundaries are granularity-dependent rather than objective. Several dialogue segmentation datasets report IAA in their original publications (e.g., DialSeg711 [6], SuperDialseg [7]), often showing only moderate agreement even under controlled annotation protocols. However, these statistics are reported using heterogeneous definitions and units (e.g., boundary placement, segment overlap, or task-level agreement), making direct comparison across datasets difficult. Because the present work focuses on evaluation behavior under fixed annotation schemes rather than on the reliability of those schemes themselves, we do not re-analyze annotator agreement here. Integrating IAA into a unified analysis of granularity effects remains an important direction for future work.

Additional directions include evaluating published dialogue segmentation methods under the proposed evaluation objective to confirm that the observed granularity effects are method-agnostic, and exploring alternative boundary selection rules that adapt boundary density to application constraints rather than annotation conventions. Further study is also needed to relate boundary density preferences to specific downstream uses such as summarization, retrieval, or conversational memory. Finally, future work could replace discrete boundary representations with continuous or multi-scale topic representations (e.g., topic clouds) that better capture overlapping, gradual, or revisited conversational structure.

11 Conclusion

Dialogue topic segmentation does not admit a single ground truth. Exact boundary matching conflates detection correctness with segmentation granularity and can obscure coherent segmentation behavior under differing annotation granularity. By separating boundary scoring from boundary selection and evaluating boundary density and segment coherence alongside tolerant boundary accuracy, this work provides a principled basis for diagnosing evaluation failures across datasets.

A central empirical finding is that boundary-based metrics can be dominated by boundary density alignment: even simple non-semantic baselines achieve competitive F1 and W-F1 scores when their output boundary rate matches the gold annotation. For example, we observe that the same published method (CSM) occupies different granularity regimes across datasets, a pattern consistent with differences in annotation density rather than changes in boundary detection quality. This underscores that boundary accuracy alone is insufficient when annotation granularity varies. For example, a targeted audit of runnable SuperDialseg method comparisons shows that higher boundary accuracy is consistently accompanied by shifts toward more aggressive segmentation regimes, rather than by improved boundary placement.

The proposed separation is intentionally simple. Its value lies in revealing systematic evaluation failures that arise when granularity is ignored. Our goal is not to claim model superiority, but to motivate reporting practices that make boundary density and segment coherence explicit. We expect these diagnostics to be useful both for benchmarking research systems and for analyzing and configuring segmentation choices in practice.

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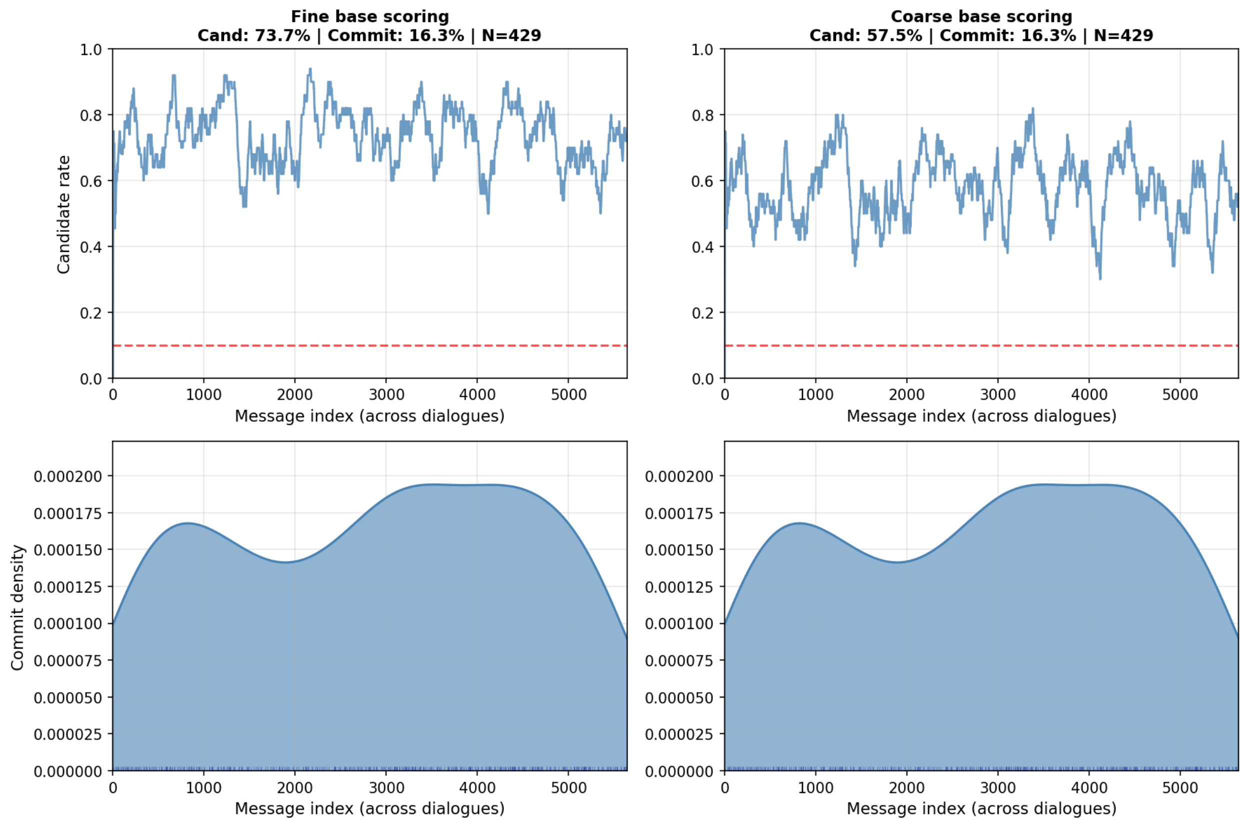

Figure 2: **Adaptive commitment enforces invariant boundary distributions across scoring granularities (DialSeg711).** *Top row* shows the raw candidate boundary rate produced by the same base scorer under fine (left) and coarse (right) candidate thresholds, resulting in substantially different candidate frequencies (73.7% vs. 57.5%). The dashed line indicates the target commit rate specified to the controller. *Bottom row* shows the resulting distribution of committed boundaries over message index after adaptive thresholding. Despite large differences in candidate generation, adaptive commitment converges to nearly identical boundary distributions and total commit counts (16.3% commit rate, N=429 in both cases). This invariance is *not* the result of score rescaling or post-hoc normalization: the underlying scoring function is unchanged, and the effect emerges from online evidence accumulation and commit gating, reflecting successful decoupling of boundary selection from boundary scoring.